

Array Partition

arr → 5, 7, 2, 1, 8, 3, (4) → last element {pivot}
 → correct position

smaller < 4 < larger

can be
a possibility

1, 2, 3, 4, 5, 7, 8

→ $n \log n$ X
 → n^2 X

Time → $O(n)$
 Space → $O(1)$
 Do not sort the whole array.

→ [3 1 2] 4 [8 7 5]

i: 0 1 2 3 4 5 6

5 7 2 1 8 3 (4)

{ si: 0
ei: 6

pivot: arr[ei]
 idx: si

idx: 0 1 2 3

5 7 2 1 8 3 4

swap ⇒ arr[i], arr[idx]

↓
 arr[i] ← pivot
 idx++

(5) 7 (2) 1 8 3 4

2 [7] 5 / 8 3 4

2 1 [5] 7 / 8 3 4

2 1 [5] 7 8 / 3 4

2 1 3 (7) 8 5 (4)

[2 1 3] [4] [8 5 7]

< pivot >

at the end
 ↓
 swap ⇒ pivot, arr[idx]

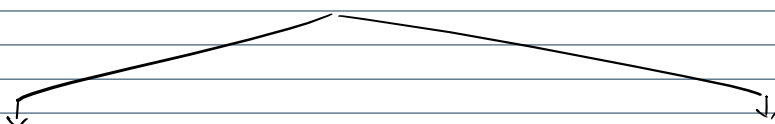
Divide and conquer

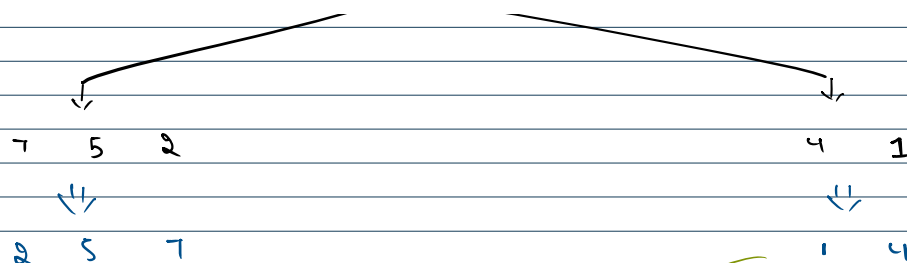
It is a powerful algorithmic strategy used to solve complex problems by breaking them down into simpler sub-problems, solving each of those independently and then combining their result to get final solⁿ.

Merge Sort

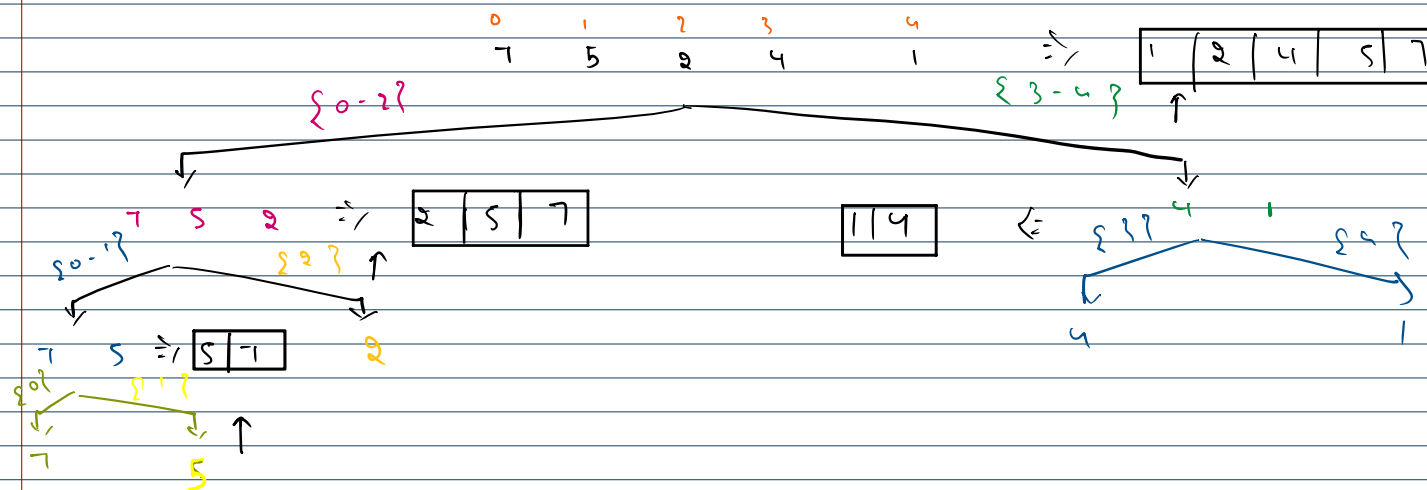
1. divide
2. conquer
3. Combine (merge)

0 1 2 3 4
 7 5 2 4 1



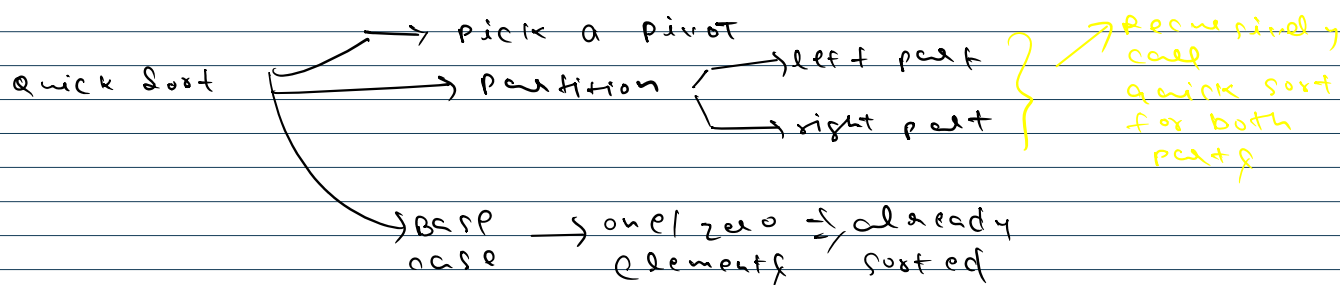


→ merge & sorted arrays

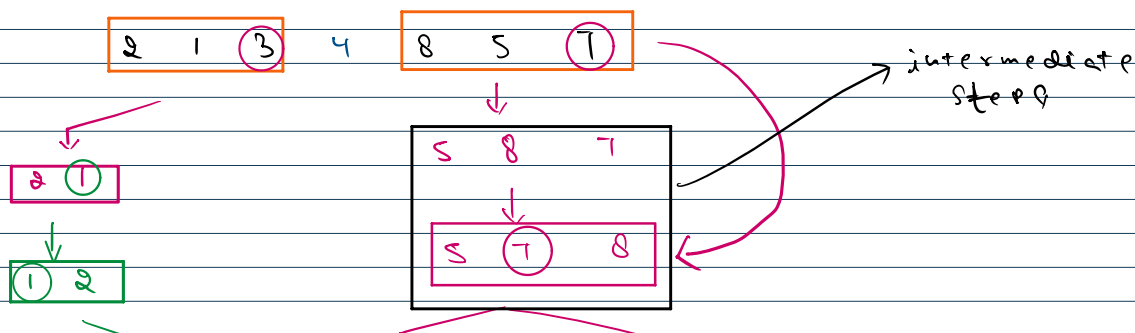


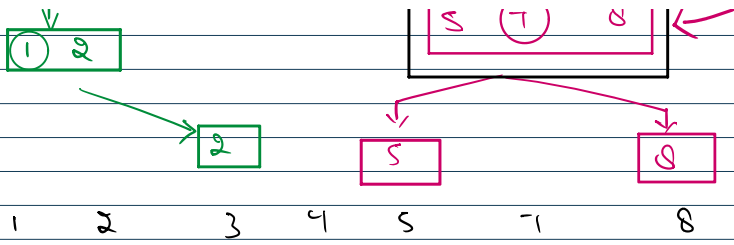
Quick Sort

Divide and Conquer Strategy.
average performance
in-place sorting. (no extra space required)

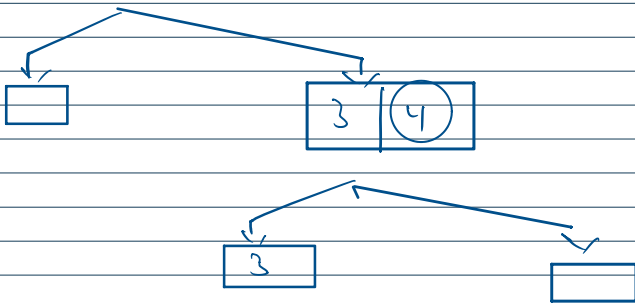
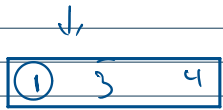
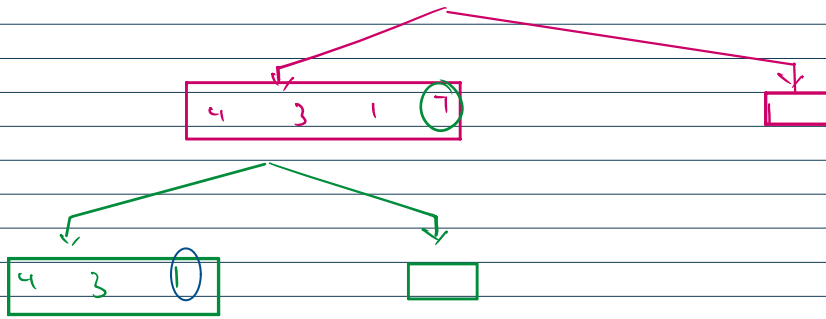


arr → 5, 7, 2, 1, 8, 3, (4)





4 3 1 7 8



1 3 4 7 8